NEP Aligned

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Data Analytics with R

Quality Laboratory Manual

Experiment No. 05

To implement a program to plot visualizations such as bar charts, pie charts, histograms, density plots and box plots

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Experiment No. 05

Title of Experiment: To implement a program to plot visualizations such as bar charts, pie charts, histograms, density plots and box plots

Aim of Experiment: To understand and implement various basic data visualization techniques in R, including bar charts, pie charts, histograms, density plots and box plots for graphical representation and analysis of data

System Requirements –

- Operating System: Windows 10 or above
- RAM: Minimum 4 GB
- Hard disk: 10 GB free space
- Processor: 2.33 GHz or higher

Software/s Needed for Experiment

- R programming language (latest stable version)
- RStudio IDE
- Spreadsheet software (Microsoft Excel/ Libre Office)
- Built-in R graphics package
- Internet **Connection (for dataset access)/ CSV dataset file**

Experiment Objectives:

- To understand the importance of data visualization in analytics
- To create bar charts for categorical data representation
- To plot pie charts to show proportion distribution
- To generate histograms to analyze frequency distribution
- To create density plots for continuous variable distribution
- To use box plots to detect spread and outliers
- To interpret graphical outputs meaningfully

Experiment Outcomes:

After completing the experiment, students will be able to –

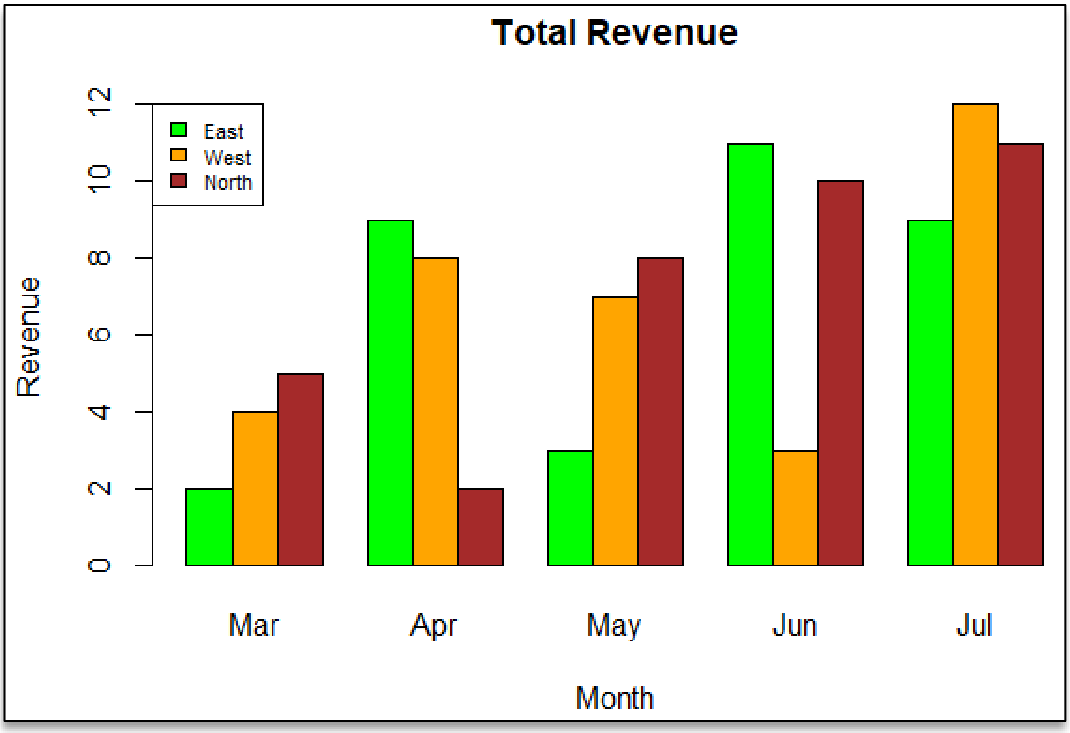
- Visualize data using different graphical methods
- Select appropriate plots based on data type
- Interpret patterns, trends and outliers
- Use R plotting functions effectively
- Apply visualization techniques in real-world analytics problems

Theory:
Data visualization is the graphical representation of information and data using visual elements such as charts, graphs, and plots. In data analytics, visualization plays a critical role in exploring datasets, identifying trends, detecting anomalies, and communicating insights effectively. Graphical representations simplify complex data structures and allow analysts to derive meaningful interpretations quickly.
R provides powerful built-in visualization tools that allow users to generate various types of plots with minimal coding effort. Visualization techniques differ based on whether the data is categorical or numerical.

Bar Chart
A bar chart is used to represent categorical data using rectangular bars, where the length or height of each bar corresponds to the value it represents. It is particularly useful when comparing quantities across different categories.

Bar charts help in:
- Comparing discrete categories
- Visualizing frequency counts
- Representing survey results

In R, the barplot() function is commonly used.

Example:
```
marks = c(75, 80, 65, 90)
names(marks) = c("A", "B", "C", "D")
barplot(marks, col="blue", main="Student Marks")
```


Pie Chart

A pie chart represents data as slices of a circle, where each slice corresponds to a proportion of the total. It is suitable for displaying percentage distributions.

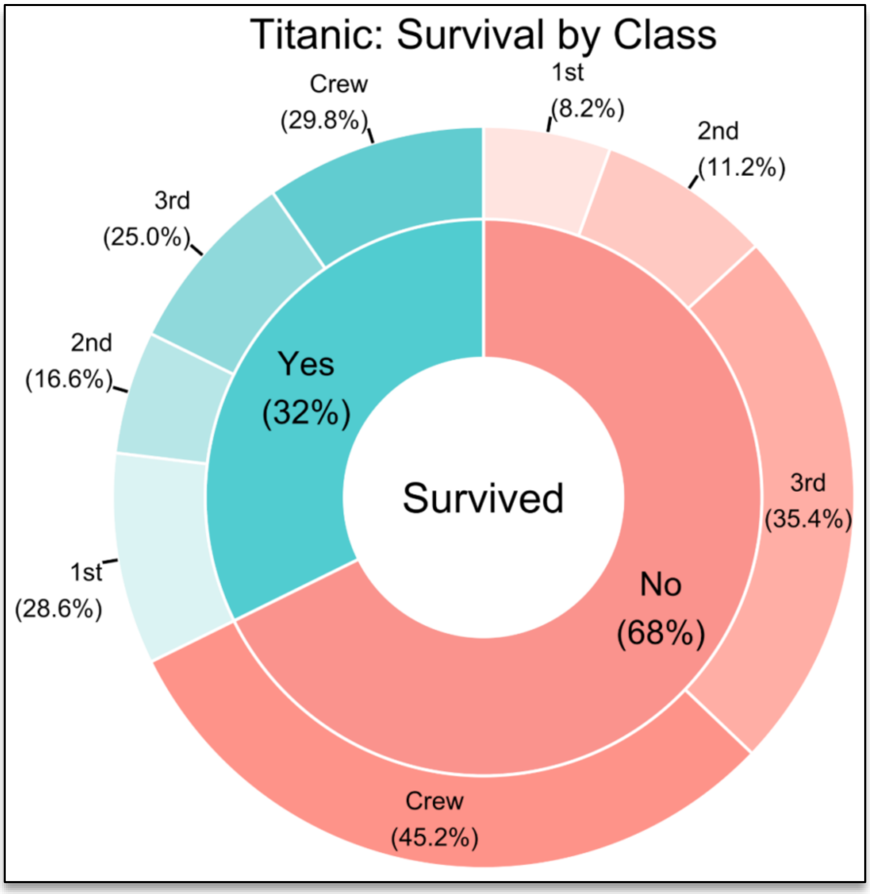
Pie charts help in:

- Showing relative proportions
- Visualizing categorical percentage share

In R, the pie() function is used.

Example:

```
sales = c(40, 30, 20, 10)
labels = c("Product A", "Product B", "Product C", "Product D")
pie(sales, labels=labels, main="Sales Distribution")
```



Histogram

A histogram is used to represent the frequency distribution of continuous numerical data. It groups data into bins (intervals) and shows how many observations fall within each bin.

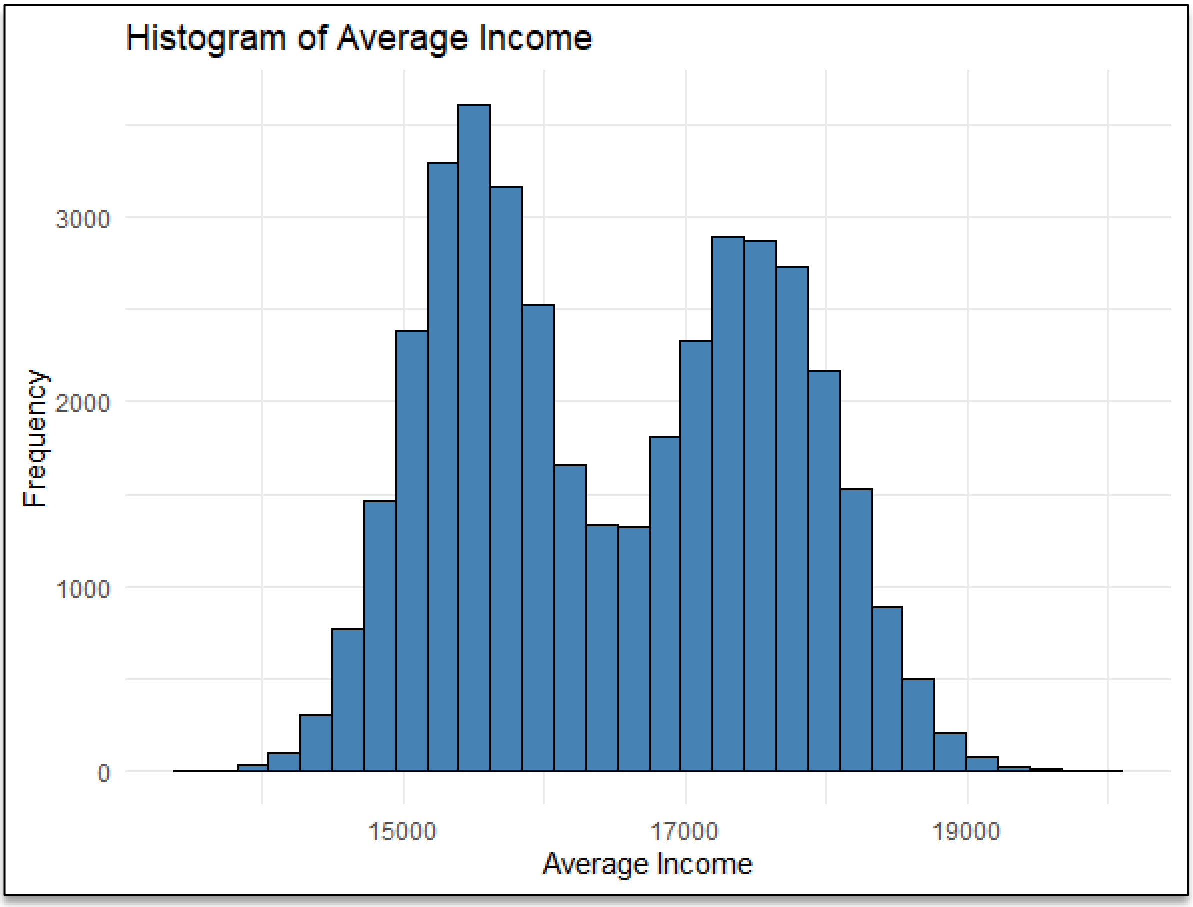
Histograms help in:

- Understanding data distribution

- Identifying skewness
- Detecting spread and frequency patterns

In R, the hist() function is used.

Example:
```
marks = c(45, 60, 75, 80, 90, 55)
hist(marks, col="green", main="Histogram of Marks")
```

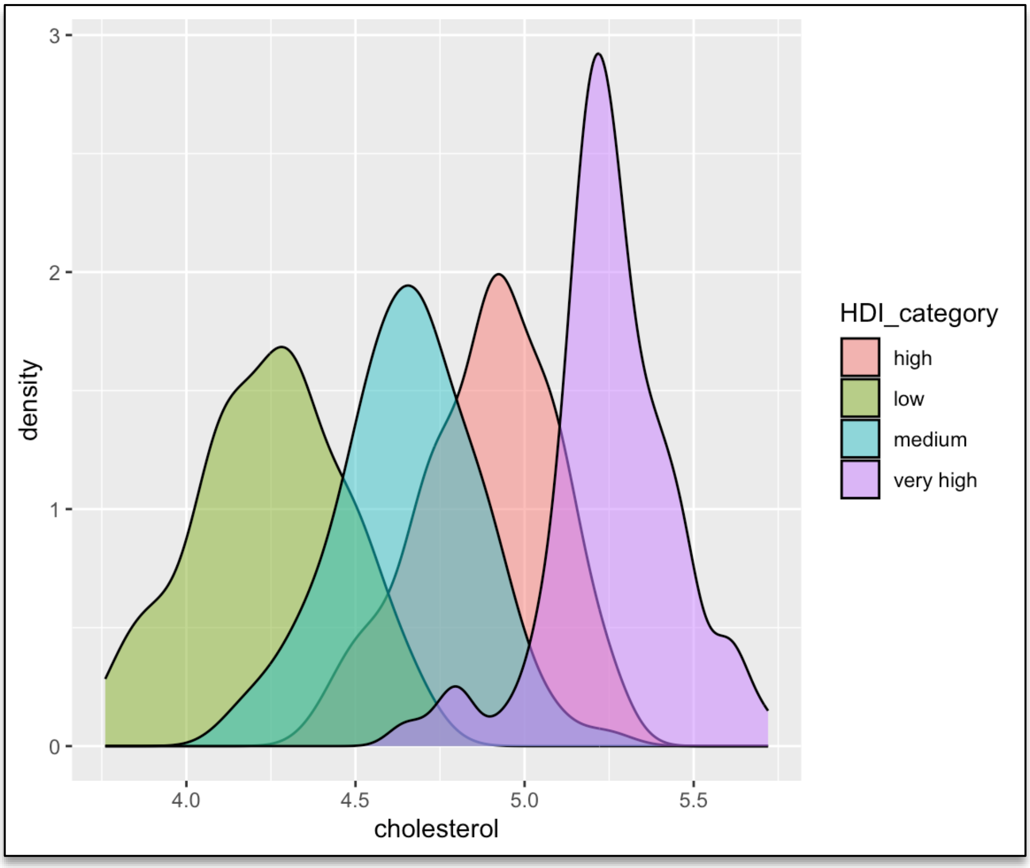


Density Plot
A density plot is a smoothed version of a histogram that shows the probability density of a continuous variable. It provides a better understanding of the distribution shape without being affected by bin size.

Density plots help in:
- Comparing distributions
- Observing peaks and spread
- Identifying skewness

In R, density is calculated using density() and plotted using plot().

Example:
```
plot(density(marks), main="Density Plot of Marks")
```

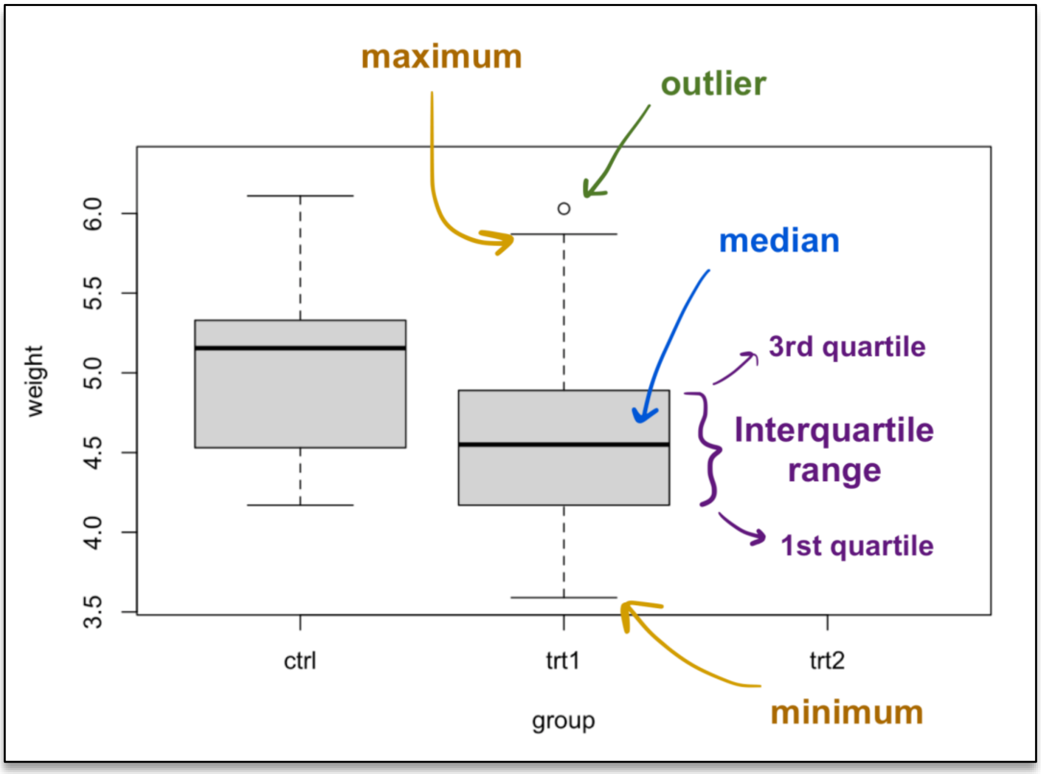


Box Plot
A box plot (box-and-whisker plot) summarizes data using five-number summary:

- Minimum
- First Quartile (Q1)
- Median
- Third Quartile (Q3)
- Maximum

It is particularly useful for identifying outliers and understanding variability.
In R, boxplot() function is used.

Example:
```
boxplot(marks, col="orange", main="Box Plot of Marks")
```



Task to Perform

Basic Data Visualization using R – Visualizing student performance data using R

You are given marks of 50 students in a data analytics test

```
marks = c(45, 60, 75, 80, 90, 55, 70, 65, 85, 50,
          72, 68, 77, 83, 91, 58, 63, 74, 88, 52,
          66, 79, 81, 69, 73, 59, 62, 84, 87, 53,
          71, 76, 82, 64, 67, 54, 61, 78, 86, 57,
          92, 48, 56, 89, 93, 47, 95, 49, 94, 51)
```

Perform the following tasks: **(Add proper titles, axis labels and suitable colors – no gray scale charts)**

1. Create a Bar Chart for student marks
2. Create a Pie Chart showing proportion of marks
3. Plot a Histogram for the mark distribution
4. Plot a Density Plot for marks
5. Create a Box Plot to analyze spread and outliers

Observations:

- Bar charts clearly represented categorical comparisons
- Pie charts effectively displayed proportion distributions
- Histograms showed frequency distribution and data spread
- Density plots provided smoother visualization of distribution
- Box plots helped in identifying median and possible outliers
- Different plots are suitable for different types of data

Conclusion:

The experiment successfully demonstrated various basic data visualization techniques using R. students learned to generate and interpret bar charts, pie charts, histograms, density plots and box plots. Visualization plays a crucial role in exploratory data analysis and is a fundamental skill required in data analytics and machine learning applications

Expected Oral Questions:

1) What is data visualization?
2) When should we use a bar chart?
3) Difference between histogram and bar chart
4) What does a box plot represent?
5) What is skewness?
6) Why are density plots preferred over histograms sometimes?
7) What is the five-number summary?
8) Which plot is suitable for detecting outliers?

FAQs in Interview:

1) Why is visualization important in analytics?
2) What is the difference between histogram and density plot?
3) When should a pie chart be avoided?
4) How does a box plot detect outliers?
5) Which R library is commonly used for advanced visualizations?